

BUS33-3

Corporate Finance

Invest, Finance, Return

Making capital decisions for a company, not just reading it.

Albert School — 22 hours

Contents

1	Time value of money	2
2	Net Present Value and the investment decision	2
3	Internal Rate of Return	3
4	Cost of equity via CAPM	3
5	Cost of debt	4
6	Weighted Average Cost of Capital	4
7	Capital structure	5
8	Payout policy	5
9	Value-creation metrics	6
10	Integrated investment case	6
10.1	Critiquing a capital allocation proposal	6

1 Time value of money

A sum of money available today is not equivalent to the same nominal sum available later: money today can be invested to earn a return, so it is worth more. The time value of money is the machinery that makes cash flows occurring at different dates comparable.

Definition 1 (Future and present value) At a periodic rate r , a sum C invested for n periods grows by *compound interest* to its *future value*

$$FV = C(1 + r)^n.$$

Conversely, the *present value* of a sum C_n received n periods from now is obtained by *discounting*:

$$PV = \frac{C_n}{(1 + r)^n}.$$

The factor $1/(1 + r)^n$ is the *discount factor*.

For a stream of cash flows $C_1, C_2, \dots, C_n$ occurring at the end of periods 1 through n , present values add:

$$PV = \sum_{t=1}^n \frac{C_t}{(1 + r)^t}.$$

Two standard streams have closed forms. A *perpetuity* paying C each period forever has $PV = C/r$. A *growing perpetuity* whose payment grows at rate $g < r$ has $PV = C/(r - g)$. An *annuity* paying C for n periods has

$$PV = \frac{C}{r} \left(1 - \frac{1}{(1 + r)^n} \right).$$

The discount rate r must match the cash flows: nominal rate with nominal cash flows, and the same periodicity (a monthly stream is discounted at a monthly rate). Every later chapter — NPV, the cost of capital, valuation — is an application of this single principle.

2 Net Present Value and the investment decision

An investment project is a set of dated cash flows: an outflow now (and possibly later) in exchange for inflows in the future.

Definition 2 (Net Present Value) The *Net Present Value* of a project with cash flows $C_0, C_1, \dots, C_n$ (with C_0 typically negative — the initial outlay) discounted at rate r is

$$\text{NPV} = \sum_{t=0}^n \frac{C_t}{(1+r)^t}.$$

NPV measures the value created by the project today, after compensating investors for the time value of money and risk through the discount rate.

Proposition 3 (NPV rule) Accept a project if $\text{NPV} > 0$; reject it if $\text{NPV} < 0$. Among mutually exclusive projects, choose the one with the highest NPV.

A positive NPV means the project returns more than the cost of the capital it ties up, so it increases the value of the firm. The discount rate used is the cost of capital (Chapters 4–6), reflecting the risk of the project’s cash flows.

3 Internal Rate of Return

Definition 4 (Internal Rate of Return) The *Internal Rate of Return* of a project is the discount rate that makes its NPV zero:

$$\sum_{t=0}^n \frac{C_t}{(1 + \text{IRR})^t} = 0.$$

For a conventional project (one sign change: an outflow followed by inflows), the IRR rule — accept if IRR exceeds the cost of capital — agrees with the NPV rule, because NPV is positive precisely when the discount rate is below the IRR.

The two rules can conflict:

- **Scale.** IRR is a rate and ignores project size; a small project can have a higher IRR but a lower NPV than a large one. For mutually exclusive projects of different scale, NPV ranks correctly.
- **Timing.** Projects with different cash-flow timing can rank differently under IRR and NPV; the ranking under IRR can even reverse as the cost of capital changes.
- **Multiple or no IRR.** A project whose cash flows change sign more than once can have several IRRs or none, leaving the IRR rule undefined.

NPV is the superior rule because it measures value added directly, in currency, and at the relevant cost of capital. IRR remains useful as a communicable summary of a project’s return that can be compared against the cost of capital without restating the discount rate.

4 Cost of equity via CAPM

Equity holders bear residual risk and require a return for it. The Capital Asset Pricing Model expresses that required return as a function of systematic risk.

Definition 5 (CAPM) The *cost of equity* is

$$r_e = r_f + \beta(r_m - r_f),$$

where r_f is the *risk-free rate*, β is the stock's sensitivity to market movements, and $r_m - r_f$ is the *equity risk premium* — the excess return investors demand for holding the market over the risk-free asset.

Each input requires judgment:

- **Risk-free rate.** Proxied by a government bond yield; the maturity is chosen to match the horizon of the cash flows being discounted.
- **Beta.** Estimated by regressing the stock's returns on market returns, or sourced from a provider. Estimates depend on the time window, the return frequency, and the market index chosen, and may be adjusted toward 1.
- **Equity risk premium.** Estimated from long-run historical excess returns or implied from current prices; the two approaches can give materially different figures, so the choice must be justified.

5 Cost of debt

Definition 6 (Cost of debt) The *pre-tax cost of debt* r_d is the return a lender requires today — estimated as the *yield to maturity* on the firm's outstanding bonds, or as the risk-free rate plus a *credit spread* reflecting default risk.

Because interest is tax-deductible, debt's true cost to the firm is lower than the rate paid. At a corporate tax rate τ , the *after-tax cost of debt* is

$$r_d(1 - \tau).$$

Definition 7 (Debt tax shield) The *tax shield* is the tax saved because interest is deductible: on interest expense I , the saving is τI per period.

The tax shield matters because it makes debt financing cheaper than its stated rate, and is the mechanism through which capital structure can affect firm value (Chapter 7). The relevant cost of debt for decisions is the after-tax, forward-looking rate — the rate the firm would pay to borrow today, not the historical coupon on old debt.

6 Weighted Average Cost of Capital

A firm financed by both equity and debt must earn enough to satisfy both providers. The WACC is the blended required return, weighted by the capital structure.

Definition 8 (WACC) With market values of equity E and debt D , total $V = E + D$, the *Weighted Average Cost of Capital* is

$$\text{WACC} = \frac{E}{V}r_e + \frac{D}{V}r_d(1 - \tau),$$

where r_e is the cost of equity, r_d the pre-tax cost of debt, and τ the corporate tax rate.

The weights are target (or market-value) weights, not book weights, because they represent the cost of raising capital today. WACC is the discount rate for a project of the same risk as the firm as a whole.

The effect of leverage on WACC is not monotonic. Replacing equity with cheaper after-tax debt lowers the average at first; but more debt raises the riskiness of both equity and debt, pushing r_e and r_d up. A project earning a return above the firm's WACC creates value; one earning below it destroys value — the link to Chapter 9.

7 Capital structure

Capital structure is the mix of debt and equity used to finance the firm.

Proposition 9 (Modigliani–Miller irrelevance) In a frictionless market — no taxes, no bankruptcy costs, no information asymmetry — the total value of a firm is independent of how it is financed. Splitting the cash flows between debt and equity does not change their total.

The proposition is a benchmark, not a description of reality: it isolates the frictions that actually make capital structure matter.

- **Taxes.** Interest deductibility creates a tax shield (Chapter 5) that adds value as leverage rises — pushing toward more debt.
- **Distress costs.** Higher leverage raises the probability of financial distress and bankruptcy, which carries direct and indirect costs — pushing toward less debt.
- **Agency costs.** Debt can discipline managers by committing free cash flow to interest, but excessive debt creates conflicts between shareholders and creditors.

The trade-off between the tax shield and distress costs implies a target leverage range. Diagnosing a firm as *over-leveraged* (distress costs and constrained investment dominate) or *under-leveraged* (an unused tax shield, possibly slack discipline) means locating it relative to that range, using leverage and coverage ratios from BUS13-3 and BUS23-3.

8 Payout policy

Once a firm generates cash beyond its profitable investment needs, it must decide how much to return to shareholders and by what mechanism.

Definition 10 (Payout mechanisms)

- A *dividend* is a cash distribution per share, typically regular.
- A *share buyback* (repurchase) returns cash by buying the firm's own shares, reducing the share count.

The two differ in:

- **Tax.** Depending on the jurisdiction and the shareholder, dividends and capital gains from buybacks are taxed differently, which can favour one mechanism.
- **Signalling.** A raised regular dividend signals confidence in durable future cash flows and is costly to cut, so it is a strong commitment; buybacks are more flexible and signal less.

The *retention vs distribution* decision is the prior question: cash should be retained only if the firm has investments earning above its cost of capital (positive NPV); otherwise it should be

distributed. A payout-policy recommendation follows from the firm's growth opportunities, the stability of its cash generation, and its shareholder base's tax position and preferences.

9 Value-creation metrics

A firm creates value only when it earns more on its invested capital than that capital costs.

Definition 11 (ROIC vs WACC test) With *Return on Invested Capital* $ROIC = NOPAT / \text{invested capital}$ (from BUS23-3), the firm *creates value* when $ROIC > WACC$, is value-neutral when $ROIC = WACC$, and *destroys value* when $ROIC < WACC$.

Definition 12 (Economic Value Added) *Economic Value Added* expresses the same test as a currency amount:

$$EVA = (ROIC - WACC) \times \text{invested capital} = NOPAT - WACC \times \text{invested capital} .$$

Positive EVA means value created over the period; negative EVA means value destroyed.

Unlike accounting profit, EVA charges the firm for *all* capital employed, including equity, at its cost. A firm can report positive net income yet a negative EVA if its return does not clear its cost of capital. Computing ROIC and EVA for a real company turns the cost of capital from Chapters 4–6 into a verdict on whether the company is worth more or less for having operated.

10 Integrated investment case

The three corporate-finance decisions — *invest*, *finance*, *return* — are made together for one company, using accounting-adjusted inputs from BUS23-3 (normalized earnings, free cash flow, ROIC).

- **Invest.** Build the project or strategy as dated free cash flows and evaluate it with NPV at the appropriate discount rate, cross-checked with IRR (Chapters 2–3).
- **Finance.** Choose a financing mix and compute its WACC, assessing how the proposed leverage moves the cost of capital and where the firm sits relative to its target structure (Chapters 5–7).
- **Return.** Justify a payout policy for the cash not absorbed by positive- NPV investment, consistent with the firm's growth and cash generation (Chapter 8).

The integrated case is coherent only if the same inputs and risk assumptions run through all three decisions.

10.1 Critiquing a capital allocation proposal

A finished case must also be challenged. A disciplined critique:

- identifies which *assumptions drive the NPV* — typically the growth rate, margins, terminal value, and discount rate — by testing how the conclusion moves when each is varied;
- locates where the *cost of capital estimate is weakest* — the equity risk premium, beta, and the target weights all rest on judgment (Chapters 4–6);
- states *what the model cannot capture* — optionality, strategic interactions, and risks not reflected in the discounted cash flows.

The recommendation is defended not by its point estimate but by knowing which inputs it depends on and how far it survives their plausible range.